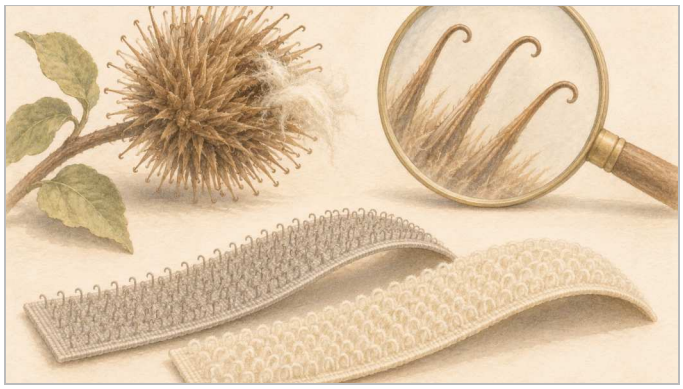
How Velcro Works

01 Where did the idea come from?



A burr's spines end in tiny hooks.

Look closer. Find the tiny hooks at the tips of the burr's spines.

A little explanation. A burr. In 1941, a Swiss engineer came home from a walk with burdock burrs stuck to his socks and his dog's fur. Under a microscope, each spine ended in a tiny hook. Burrs cling on purpose: animals carry their seeds to new places.

Try this. After a walk outdoors, look closely at your socks. Did any seeds hitch a ride? Look at one with a magnifying glass for hooks.

02 How do you make a tiny hook?



Stiff hooks above, soft loops below.

Look closer. Compare the stiff hooks on top with the soft loops below.

A little explanation. You start with a loop. The first hooks were cotton and wore out. Nylon worked better: it was woven into loops, heated so the loops kept their shape, then a cutter snipped one side of each loop to leave a hook.

Try this. Bend a pipe cleaner into a small loop. Open one side a little so it becomes a hook. Can your hook catch a loop of yarn?

03 Why does a peel open so easily?



A peel opens just one thin line at a time.

Look closer. Look at how thin the line is where the strip is letting go.

A little explanation. A sideways pull, called shear, makes every hook hold at once. A peel opens only one thin line of hooks at a time. The rrrip is hooks springing free, one row after another.

Try this. Press two strips together. Slide them apart sideways, then peel one corner. Which one takes less push? Listen to each.

04 Why do old strips lose their grip?



Lint clogs the hooks; a comb lifts it out.

Look closer. Find the fuzz in the hooks and the clean part next to it.

A little explanation. Lint and fluff pack into the hooks, so they cannot reach the loops. A fine comb can lift the fuzz out, and the grip comes back.

Try this. Find a fuzzy strip on an old shoe or bag. Comb it gently with an old comb or toothbrush, then test the grip again.